



CVL2001: Climate Change and Adaptation Section B

Lecture 5: Climate Impacts, Thresholds and Abrupt Change

Dessler, A. E., Introduction to Modern Climate Change, Chapter 9: Sections
9.3–9.5

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Quick Recap: Wayanad Case Study

Last lecture:

- Climate change intensified the rainfall hazard.
- Local conditions shaped slope failure.
- Exposure and vulnerability shaped the loss.

Today:

- What happens when a system crosses its capacity or threshold?

Learning outcome:

1. Explain why climate impacts can be nonlinear.
2. Evaluate how adaptation reduces damage while introducing costs.
3. Compare the adaptability of managed and unmanaged systems.
4. Describe how ecosystem services connect natural systems to human welfare.

From Physical Change to Human Welfare

- Physical climate change does not automatically translate into an equal amount of harm.
- Changes in temperature, rainfall, sea level and ocean chemistry
 - ✓ affect exposed people and ecosystems
 - ✓ may push systems beyond their coping capacity
 - ✓ result in social, ecological and economic damage
- A small climate shift can cause a large impact.

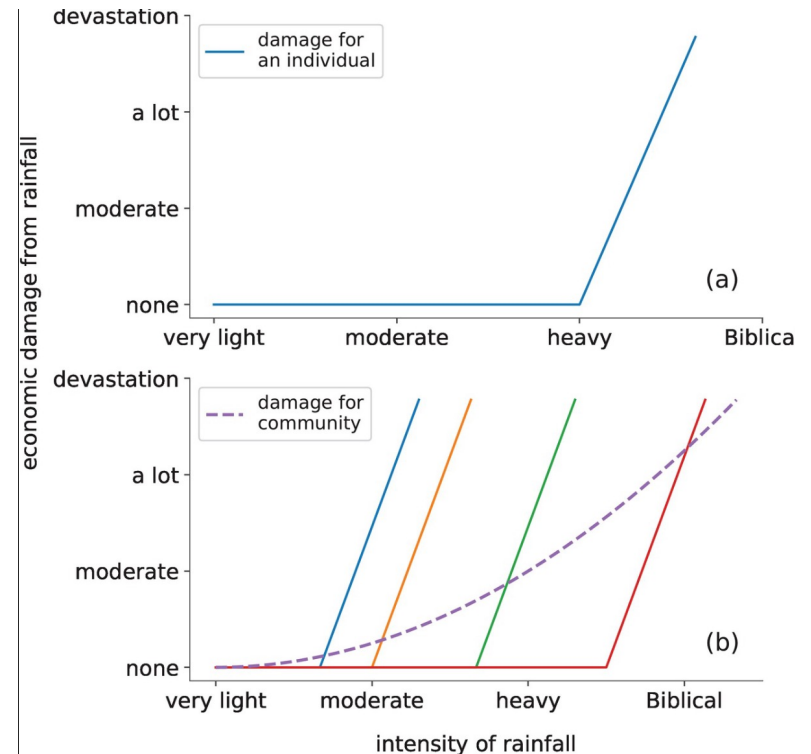


Image source:

https://upload.wikimedia.org/wikipedia/commons/4/4e/Majestic_View_of_Srisailem_Dam_Spillway_Overflow_During_Monsoon.jpg?utm_source=commons.wikimedia.org&utm_campaign=index&utm_content=thumbnail_unscaled&_=20250709053741

Impacts Need Not Rise Smoothly

- **Below a threshold:** a system may cope with little visible damage.
- **Near the threshold:** a small additional change can exhaust capacity.
- **Beyond the threshold:** damage can rise much faster than the climate variable.



Dessler, A. E., Introduction to Modern Climate Change, Chapter 9, Section 9.3.1 and Figure 9.4.

One Household Can Have a Sharp Threshold

- Rainfall remains below protective capacity: little or no flood damage
- Water rises above the protective level: the house floods and appears suddenly
- The response is nonlinear at the individual scale.



A)https://upload.wikimedia.org/wikipedia/commons/thumb/c/c6/FEMA_-_35886_-_Flooded_home_in_Missouri.jpg/1920px-FEMA_-_35886_-_Flooded_home_in_Missouri.jpg?utm_source=commons.wikimedia.org&utm_campaign=index&utm_content=thumbnail&_20091014201608

A Community Has Many Thresholds

- Households, roads and services do not fail at the same point.
- Different thresholds smooth the community response.
- But as more thresholds are crossed, total damage curves upward.

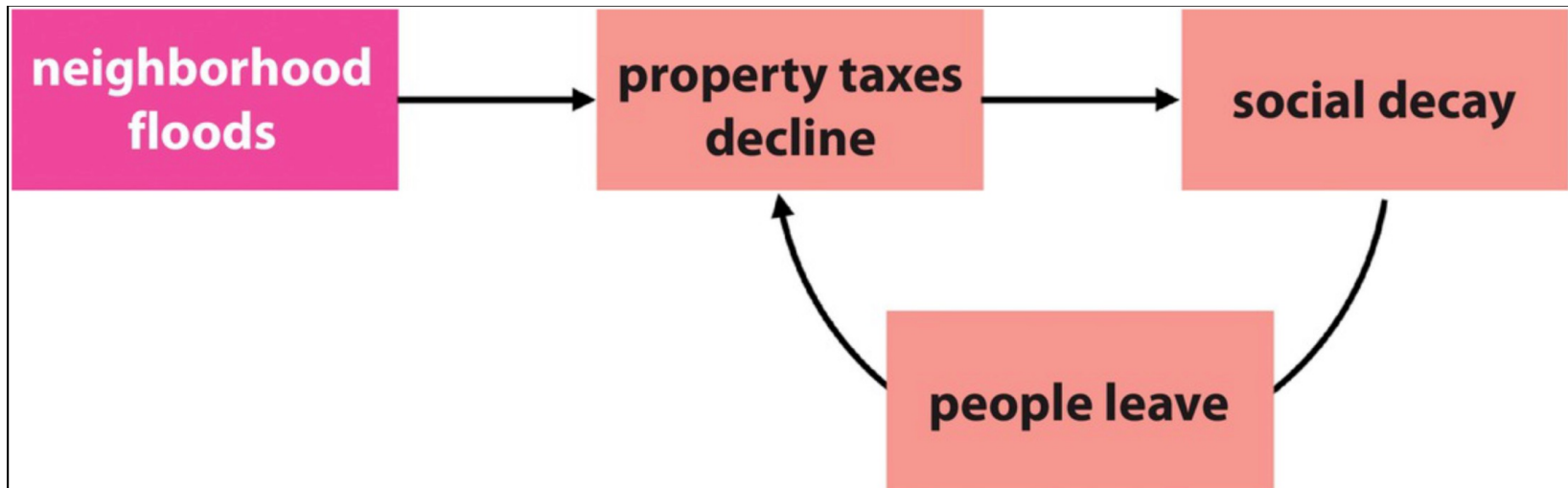


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One Impact Can Trigger Another

Flooding

- property values fall
- property-tax revenue falls
- public services weaken
- residents and firms leave
- the local economy weakens further



Adaptation Changes the Consequences

- Adaptation does not stop the climate from changing.
- It changes exposure, sensitivity or protective capacity so that a given climate hazard causes less harm.
- Humans can modify some buildings and infrastructure—but not at zero cost.



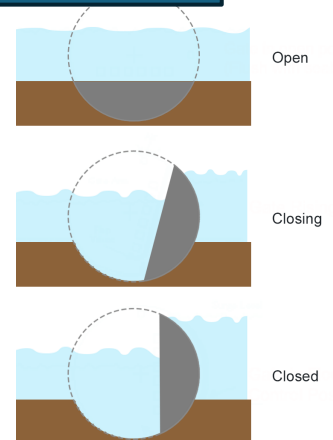
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Adaptation Itself Can Be Expensive

- Individual scale: raise a house above expected flood levels
- Community scale: build large coastal barriers
- The decision is not “cost or no cost.”
- It is adaptation cost versus avoided and residual damage.



Thames Barrier, London



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https://www.environmentandsociety.org/sites/default/files/styles/esp_image_small/public/thumbnails/image/image_2-_thames_barrier_-_simple_operation_diagram.png?itok=7kblMK4C

Q)Why can a small increase in extreme rainfall cause a disproportionately large increase in damage?

- A. Climate models always underestimate rainfall
- B. Damage may accelerate after a protective threshold is exceeded
- C. Rainfall and economic damage are unrelated
- D. Every household fails at exactly the same rainfall level

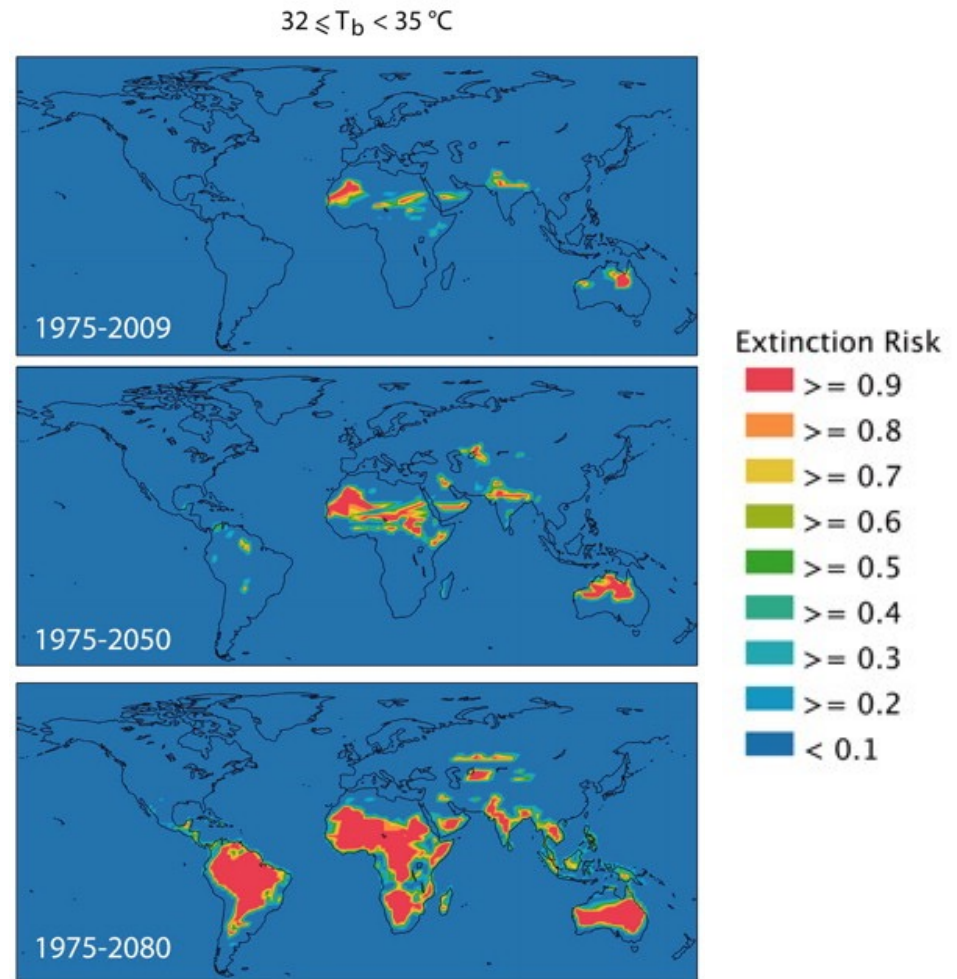
Managed and Unmanaged Systems

- **Managed human systems**
Buildings, roads and cities can be redesigned, protected, repaired or relocated.
- **Unmanaged natural systems**
Coral reefs, forests and wildlife must adjust through migration or biological adaptation.
- When climate changes too quickly, natural systems may not adjust fast enough.



Species Respond at Different Rates: Lizard Survival

- Warming shifts where species can live.
- Some adapt or migrate; others cannot.
These unequal responses disrupt food webs and pollination.
- Ecosystems can change long before species disappear.



Sinervo, Barry, et al. "Erosion of lizard diversity by climate change and altered thermal niches." *Science* 328.5980 (2010): 894-899.

Ecosystems provide services people depend on

- **Mangroves**
→ reduce erosion and storm-surge impacts
- **Pollinators**
→ support crop production
- When ecosystems are lost, society must replace these services or bear greater losses.



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Mangroves in Odisha reduced the impact of Cyclone Dana 2024

Impacts Are Already Occurring

- Climate impacts are not confined to 2050 or 2100.
- Communities already pay through:
 - disaster losses
 - repair and protection costs
 - health burdens
 - disrupted work and services

Q) Can you identify a recent extreme event in India whose likelihood, intensity or impacts may have been influenced by climate change?

Abrupt Climate Change

- Abrupt climate change is a large and rapid shift in part of the climate system that occurs over years or decades, rather than centuries.
- A gradual change in climate forcing can push the climate system across a critical threshold.
- Once crossed, the system may shift:
 - Rapidly
 - Substantially
 - On human timescales
 - With limited possibility of reversal

Why Do Abrupt Changes Matter?

Potential Abrupt Climate Changes

- Abrupt collapse of major ocean circulation
- Rapid regional climate shifts
- Sudden reorganization of monsoon patterns
- Rapid ice-shelf collapse that accelerates ice loss



<https://www.intelligentliving.co/wp-content/uploads/2020/02/permafrost-melting-in-the-arctic-region-of-high-res-stock-photography-579215990-1548678927.jpg>

These events are uncertain, but their consequences could be severe and difficult to manage.

Key Takeaways

- Small climate changes can produce disproportionately large impacts.
- Adaptation reduces damage, but cannot eliminate all risk.
- Losing ecosystem services creates human and economic costs.
- Climate thresholds may trigger rapid, difficult-to-reverse changes.

Both action and inaction have costs, but inaction leaves damages unmanaged.